

1. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        // Display prompt
        printf("myshell> ");
        fflush(stdout);

        // Read user input
        if (fgets(input, sizeof(input), stdin) == NULL) {
            printf("\n");
            break;
        }

        // Remove newline
        input[strcspn(input, "\n")] = '\0';

        // Handle exit condition
        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        // Handle empty input
        if (strlen(input) == 0) {
            continue;
        }

        // Display entered command
        printf("You entered: %s\n", input);
    }

    return 0;
}
```

```
gowtham@DESKTOP-2F03K3T:~/OS$ nano shell.c
gowtham@DESKTOP-2F03K3T:~/OS$ gcc shell.c -o shell
gowtham@DESKTOP-2F03K3T:~/OS$ ./shell
myshell> hello
You entered: hello
myshell> linux
You entered: linux
myshell> test
You entered: test
myshell> exit
Exiting shell...
```

```
gowtham@DESKTOP-2F03K3T:~/OS$ while (1) {
    printf("myshell> ");
    ...
}
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token `"myshell> "'
...: command not found
-bash: syntax error near unexpected token `}'
gowtham@DESKTOP-2F03K3T:~/OS$ |
```

```
gowtham@DESKTOP-2F03K3T:~/OS$ fgets(input, sizeof(input), stdin);
-bash: syntax error near unexpected token `input,'
gowtham@DESKTOP-2F03K3T:~/OS$ |
```

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

#define BUFFER_SIZE 100

void enable_raw_mode() {
    struct termios raw;

    tcgetattr(STDIN_FILENO, &raw);

    raw.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

void disable_raw_mode() {
    struct termios original;

    tcgetattr(STDIN_FILENO, &original);

    original.c_lflag |= (ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &original);
}

int main() {
    char buffer[BUFFER_SIZE];
    int index = 0;
    char ch;

    enable_raw_mode();

    printf("myshell> ");
    fflush(stdout);

    while (1) {

        read(STDIN_FILENO, &ch, 1);

        /* Handle Enter key */
        if (ch == '\n' || ch == '\r') {

            buffer[index] = '\0';

            printf("\n");

            if (strcmp(buffer, "exit") == 0) {
                printf("Exiting shell...\n");
                break;
            }

            if (index > 0) {
                printf("Command entered: %s\n", buffer);
            }

            index = 0;

            printf("myshell> ");
            fflush(stdout);
        }

        /* Handle Backspace */
        else if (ch == 127 || ch == 8) {

            if (index > 0) {
                index--;

                buffer[index] = '\0';

                printf("\b \b");
                fflush(stdout);
            }
        }

        /* Handle normal characters */
        else if (ch >= 32 && ch <= 126) {

            if (index < BUFFER_SIZE - 1) {

                buffer[index] = ch;
                index++;

                printf("%c", ch);
                fflush(stdout);
            }
        }
    }

    disable_raw_mode();

    return 0;
}

```

```

gowtham@DESKTOP-2F03K3T:~/OS$ nano keyboard_shell.c
gowtham@DESKTOP-2F03K3T:~/OS$ gcc keyboard_shell.c -o keyboard_shell
gowtham@DESKTOP-2F03K3T:~/OS$ ./keyboard_shell
myshell> hello
Command entered: hello
myshell> hello
Command entered: hello
myshell> hel
Command entered: hel
myshell> printf("\b \b");
Command entered: printf("\b \b");
myshell> helprintf("\b \b");
Command entered: helprintf("\b \b");
myshell> |

```

```

gowtham@DESKTOP-2F03K3T: ~$ cd OS
gowtham@DESKTOP-2F03K3T:~/OS$ nano keyboard_shell.c
gowtham@DESKTOP-2F03K3T:~/OS$ gcc keyboard_shell.c -o keyboard_shell
gowtham@DESKTOP-2F03K3T:~/OS$ ./keyboard_shell
myshell> hello
Command entered: hello
myshell> hello
Command entered: hello
myshell> hel
Command entered: hel
myshell> printf("\b \b");
Command entered: printf("\b \b");
myshell> helprintf("\b \b");
Command entered: helprintf("\b \b");
myshell> if (ch == '\n' || ch == '\r')
Command entered: if (ch == '\n' || ch == '\r')
myshell> User presses Enter
Command entered: User presses Enter
myshell>
Command entered:
myshell> '\n' detected
Command entered: '\n' detected
myshell>
Command entered:
myshell> buffer[index] = '\0'
Command entered: buffer[index] = '\0'
myshell>
Command entered:
myshell> Process command
Command entered: Process command
myshell>
Command entered:
myshell> index = 0
Command entered: index = 0
myshell>
Command entered:
myshell> Display myshell>myshell> exit
Command entered: Display myshell>myshell> exit
myshell> |

```

```

myshell>
gowtham@DESKTOP-2F03K3T:~/OS$ char buffer[BUFFER_SIZE]
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'chat' from deb ppp (2.5.2-1+1.2)
  command 'shar' from deb sharutils (1:4.15.2-13)
  command 'car' from deb ucommon-utils (7.0.1-1)
  command 'phar' from deb php8.5-cli (8.5.4-0ubuntu1.1)
  command 'phar' from deb php-cli (2:8.5+99ubuntu1)
  command 'chr' from deb chr (0.1.80-1)
  command 'chr' from deb chr-tiny (0.1.80-1)
  command 'pchar' from deb pchar (1.5-7)
See 'snap info <snapname>' for additional versions.
gowtham@DESKTOP-2F03K3T:~/OS$ |

```